

CO4-AT1-Q1:

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Course Code: CSA0652

Course Name: DAA for Complexity Analysis

Minimum Number of Perfect Squares

Problem Understanding

The problem is to find the minimum number of perfect square numbers whose sum equals the given number.

Dynamic Programming is used to store previous results and avoid repeated calculations.

Algorithm Used

Dynamic Programming

Steps:

1. Create a DP array to store minimum squares required.
2. Initialize values.
3. Check all possible square numbers.
4. Update the minimum count.
5. Return the final result.

Input

- Integer value n

Output

- Minimum number of perfect squares required

Example

For:

$$n = 12$$

Possible combinations:

$$12 = 4 + 4 + 4$$

Minimum count = 3

Complexity Analysis

Time Complexity:

$$O(n\sqrt{n})$$

Space Complexity:

$$O(n)$$

Result Analysis

The Dynamic Programming approach efficiently finds the minimum number of squares by storing previously calculated results.

Conclusion

Dynamic Programming provides an optimized solution compared to recursive methods because it avoids repeated calculations and improves execution efficiency.